NCPE Rapid Review Applicant Submission Template

Drug:	Vadadustat (Vafseo®)
Therapeutic indication:	Symptomatic Anaemia secondary to Dialysis-Dependent Chronic Kidney Disease
Applicant Company:	Sophie, Purnima & Anna
Submission checklist complete:	Yes
Applicant company representatives:	Anna Sheldon, Sophie Blaney & Purnima Dhingra
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List of Abbreviations

HIF-PHI – Hypoxia-Inducible Factor Prolyl-Hydroxylase Inhibitor

EPO – Erythropoietin

Hb – Haemoglobin

INNO2VATE – Pivotal RCT for Vadadustat versus Darbepoetin Alfa

CKD – Chronic Kidney Disease

ESKD – End Stage Kidney Disease

DD-CKD – Dialysis Dependent Chronic Kidney Disease

RBC – Red Blood Cells

ESA – Erythropoietin Stimulating Agent

ALT – Alanine Transaminase (liver enzyme)

AST – Aspartate Transferase (liver enzyme)

MACE – Major Adverse Cardiovascular Events

1. Details of the Intervention

Details of the Intervention	
1a: Regulatory status	
International non-proprietary name:	Vadadustat
Proprietary Name:	Vafseo
Therapeutic indication:	*Symptomatic Anaemia Dialysis Dependent- Chronic Kidney Disease (DD-CKD)*
Currently designated an orphan medicine by the EMA:	No
Date of marketing authorisation:	24/04/2023 Centralised Authorisation EMA Number: EMEA/H/C/005131/0000
Other EMA/HPRA/FDA approved indication(s), or licence extensions under EMA assessment (date of approval/status of assessment)	EMA licensed indications: *The treatment of symptomatic anaemia associated with chronic kidney disease in adults on chronic maintenance dialysis.* FDA licensed indications: *The treatment of anaemia due to chronic kidney disease in adults who have been receiving dialysis for at least three months.*
1b: Reimbursement status	
Requested reimbursement setting:	Community Drug Schemes (CDS)
Current reimbursement status:	Not currently reimbursed
Other reimbursed formulations:	N/A
1c: Description of drug	
Formulation (pack size)	150mg film-coated tablet (28-tablet pack)
Dose and frequency	300mg OD, titrated to maximum 600mg OD, titrated according to Hb response
Route of administration	Oral
Store in a refrigerator	No
Duration of use	*Given irreversible nature of this condition, it is expected patients continue this drug at a maintenance dose lifelong. Serum Hb measurements are required every two weeks until stable then every month. Dose titrations can be increased by increments of 150mg. The medication may be stopped if a clinically meaningful response in Hb is not achieved within 24 weeks of therapy. *****
ATC code	B03XA08
Mechanism of action	Vadadustat is a hypoxia-inducible factor prolyl-hydroxylase inhibitor which leads to increased cellular levels of hypoxia inducible factor stimulating endogenous EPO production. This leads to increased RBC production and iron mobilisation, resulting in increased Hb levels.

Mg= milligram; OD = once daily; Hb= Haemoglobin;
**As detailed in Summary of Product Characteristics, see references*

2. The Disease and Place in Therapy

2.1. Anaemia Secondary to Dialysis Dependent Chronic Kidney Disease

Dialysis Dependent Chronic Kidney Disease (DD-CKD) is characterised by irreversible damage to the glomerular filtration and endocrine functions of the kidney. Patients rely on exogenous blood filtering through dialysis in order to excrete nitrogenous waste and maintain a fluid balance. Most commonly kidney damage is caused by poorly controlled diabetes mellitus, hypertension and glomerulonephritis or is related to genetic structural abnormalities of the kidney (Kalantar-Zadeh *et al.*, 2021).Clinically it is characterised by increased protein in the urine, often complete anuria, hyperkalaemia and elevated serum creatinine(Webster *et al.*, 2017; Kalantar-Zadeh *et al.*, 2021). Kidney function is measured via an estimated glomerular filtration rate (eGFR), the rate at which blood is filtered through the functional unit of the kidney, based on serum creatinine levels and body surface area equations. End Stage Kidney Disease (ESKD) is defined as an eGFR <15ml/min/1.73m2, which is the point at which patients usually require dialysis support, this is compared to a normal value of >60 ml/min/1.73m2 (Palmer *et al.*, 2014).Given the irreversible nature, ultimately most patients require a kidney transplant, but given shortage of donors they rely on life saving dialysis in the interim .(Bello *et al.*, 2019)

Anaemia, a reduction in circulating healthy red blood cells (RBC) and thus haemoglobin and its oxygen carrying capacity, is a common complication of CKD. This is primarily due to deterioration of the endocrine functions of the kidney, iron deficiency, inflammation and accumulation of uraemic toxins which lead to decreased RBC survival (Babitt et al 2012). The Kidney Disease Improving Global Outcomes (KDIGO) guidelines define anaemia as haemoglobin value <13 g/dL in males and <12 g/dL in females (McMurray *et al.*, 2012).The prevalence of anaemia increases with declining renal function, meaning it affects the majority of patients who are dependent on dialysis. One of the main contributing factors is decreased erythropoietin (EPO) synthesis and EPO resistance, required to stimulate RBC production within the bone marrow. Additionally, haemodialysis can worsen anaemia given associated blood loss and further damage to RBCs. Anaemia is characterised by lethargy, shortness of breath, reduced exercise tolerance and reduced cognitive function. Clinical signs include skin pallor, pale conjunctiva and angular stomatitis. Anaemia can in turn worsen kidney function and increase the risk of cardiovascular events. These features ultimately lead to a diminished health related quality of life in patients with DD-CKD.

RBC transfusion is a method by to replace repleted red blood cells for the management of anaemia and is supported by oral and intravenous iron supplementation. However, transfusion is not without its risks including hypersensitivity and adverse reactions alongside difficulties managing a tight fluid balance in patients with DD-CKD (Ogieuhi *et al.*, 2024). Additionally, it requires continuous observation and monitoring of vital signs. Management of anaemia in DD-CKD is a substantial healthcare burden and is associated with increased healthcare utilisation (Ershler *et al.*, 2005). The introduction of exogenous EPO administration methods and now hypoxia-inducible factor prolyl hydroxylase inhibitors (HIF-PHI) has revolutionised management of anaemia in patients with DD-CKD, mitigating the risks associated with transfusion.

Vadadustat is an oral HIF-PHI which stimulates endogenous EPO production in the kidney and liver by mimicking the body's response to hypoxia (Gupta and Wish, 2017). This results in increased RBC production, erythropoiesis, for the management of anaemia in patients with DD-CKD. This class of drugs causes considerably lower but consistent levels of EPO compared to exogenous EPO administration. It is hypothesised that this reduces the cardiovascular side effect risk profile compared to the available comparators (Gupta and Wish, 2017). Vadadustat is taken orally once daily and titrated according to response in haemoglobin levels. The role of Vadadustat in the management of patients with DD-CKD in Ireland will be explored further in this rapid review.

2.2. Epidemiology of the disease in Ireland

As per the National Renal Office, Ireland, End-stage kidney disease (ESKD) affects approximately 0.10% of the population in Ireland, out of these patients, around 48% are dialysis dependent (Mellotte, 2024). Additionally, 90% of the patients on dialysis develop anaemia (Hashmi, Aeddula and Shaikh, 2024). Incident cases, representing new diagnoses of dialysis dependent patients who develop anaemia, account for approximately 2.2% of the population each year (Mellotte, 2024). In the first year, incident patients are considered null, since the prevalent population represents the gross number of patients. Kaballo et al (2018)highlights the mortality rate among dialysis patients is 16.5% and is used in our model. All patients within our eligible population are considered for treatment with the intervention Vadadustat, with 100% eligibility under the licensed indication.

Table 1: Population expected to be eligible for Vadadustat in Ireland

Eligible population	2025	2026	2027	2028	2029
Patients eligible for Vadadustat	1528	1591	1620	1648	1676

2.3. Clinical guidelines and standard of care in Ireland

A significant portion of ESKD patients experience anaemia. The current standard practice for treating symptomatic anaemia in adults undergoing dialysis for chronic kidney disease includes the use of Erythropoiesis-Stimulating agents (ESAs) supported by iron supplementation. There are no specific Irish guidelines for the treatment of anaemia resulting from ESKD. However, Ireland aligns its clinical practices with international standards, notably those from the UK National Institute for Health and Care Excellence (NICE) and the Kidney Disease: Improving Global Outcomes (KDIGO) organisation (Locatelli *et al.*, 2008). Under these guidelines first line management involves ruling out other causes of anaemia such as bleeding followed by establishing iron stores by serum assessment of percentage of hypochromic red cells, transferrin saturation and ferritin. Oral and intravenous replacement of iron is undertaken prior to commencing ESAs if required. Choice of ESAs should be patient centred considering their wishes to self-administer the medication or have supervised administration with steps made to ensure safe and efficient access to medication supplies. Hb levels should be maintained at 10 to 12g/dL. NICE recommends iron stores to be checked every one to three months with Hb assessment twice weekly during the induction of ESA therapy or after a change in dose. Once Hb is stabilised on a maintenance dose of ESA Hb levels should be assessed every one to three months (National Institute for Health and Care Excellence (NICE), 2021).

2.4 Place in therapy

Vadadustat represents an oral drug option for the management of anaemia in DD-CKD instead of ESAs or as an alternative treatment option for patients with anaemia resistant to ESAs (National Institute for Health and Care Excellence (NICE), 2025a). The role of ESAs is discussed further in the next section. One of the main advantages of Vadadustat is the oral preparation and ease of administration. Vadadustat does not require refrigeration either. These are seen as benefits over ESAs, as they require a subcutaneous or intravenous administration which can be painful for patients and may require administration by a healthcare professional or patient training to allow self-administration. Comparatively ESAs require refrigeration which could be considered an additional cost. Vadadustat is likely to be particularly attractive to individuals on home peritoneal dialysis who may prefer to take a tablet over self-administered injection. Ease of administration has the potential to increase medication compliance in this patient group.

Possible perceived disadvantages of Vadadustat includes increasing the pill burden of individuals with DD-CKD who are likely to have multiple other regular oral medications. Additionally given ESAs can be administered intravenously alongside haemodialysis therapy there would not be any additional injection required, however for the purpose of this rapid review, we will focus on a base case scenario where ESA comparators are administered subcutaneously. Importantly, despite choice of ESA or Vadadustat individuals with anaemia secondary to DD-CKD still require regular blood tests to assess Hb levels and iron stores. Individuals may still need iron replacement with Vadadustat so it does not complete resolve use of needles and associated pain.

2.5 Comparators

The main comparators are the currently available Erythropoietin Stimulating Agents (ESAs) which represent a broad class of recombinant human erythropoietin and its derivatives. The main mechanism of action is the stimulation of erythropoiesis by exogenous erythropoietin administration. Importantly, RCTs have confirmed an increased risk of cardiovascular events and mortality with ESAs targeted to near normal Hb levels. As a result, the target Hb level has been reduced in patients with DD-CKD. ESAs are administered subcutaneously or intravenously, with short-acting preparations taken every two to three days and long-acting preparations administered weekly. The dose is titrated according to change in Hb level with blood tests at least monthly and aiming at concentrations no higher than 12 g/dL (McMurray *et al.*, 2012). A Cochrane network meta-analysis (Chung *et al.*, 2023) importantly reports the comparative safety and efficacy between currently available ESAs is insufficient to determine a definitive choice. It is instead it is advised ESA choice should be based on individual patient and health system factors.

The two main options chosen for this comparison is based on Irish standard care and the medications which are currently reimbursed by the HSE. This includes Darbepoetin Alfa (Aranesp Pre-Filled "SureClick" Pen, Amgen Ireland Ltd) and Epoetin Beta (NeoRecormon Pre-Filled Syringe, Roche Registration GmbH). The choice of ESA in Ireland is largely based on availability of drug, preference of route of administration and physician choice. For the purpose of this comparison, we have chosen subcutaneous preparations on the assumption this is administered in the patients home and not in hospital, and the medication is dispensed from a community pharmacy.

1. ***Darbepoetin Alfa (Aranesp Pre-Filled "SureClick" Pen, Amgen Ireland Ltd):*** Darbepoetin Alfa is an analogue of human recombinant erythropoietin (r-HuEPO) with additional carbohydrate chains allowing for increased activity and a longer half-life (Egrie and Browne, 2001). In the chosen formulation it is

administered subcutaneously once per week. Clinical effectiveness has been established in comparison to r-HuEPO and placebo, as it reduces the need for RBC infusions in patients with advanced CKD (Palmer *et al.*, 2014). However, a more recent meta-analysis suggests an increased odds of stroke compared to placebo (Chung *et al.*, 2023). Darbepoetin Alfa is dosed at 0.45mcg/kg body weight in patients with CKD given subcutaneously or intravenously at a preferred once weekly dosing. A dose titration of up to 25% of starting dose every four weeks to maintain target Hb is recommended. If Hb increase >2g/dL in four weeks, then the dose is reduced by 25% (Amgen Europe, 2023a). This medication is reimbursed by the HSE under the Hight-Tech Drug Scheme and must be stored in a refrigerator at 2 to 8 °C. Darbepoetin Alfa is compared to Vadadustat in our pivotal trial, INNO2VATE (K.-U. Eckardt *et al.*, 2021), detailed further in section 3. The dosing assumptions and calculations are detailed further in section 4.

2. ***Epoetin Beta (Neo Recormon Pre-Filled Syringe, Roche Registration GmbH):*** Epoetin Beta is also an analogue of r-HuEPO and consists of an almost identical structure and sequence to endogenous EPO (Storring *et al.*, 1998). Epoetin beta has a longer half-life when administered subcutaneously meaning on cost and pharmacological grounds this route of administration is preferred (Locatelli *et al.*, 2002). Dosing is based on an initial dose of 60IU/kg body weight per week in patients with CKD. The dose is titrated every four weeks by 60IU/kg/week if Hb increase no adequate (less than 0.15g/dL increase). Once target Hb (10 to 12 g/dL) achieved the maintenance dosing phase begins which aims to half the previous weekly dose (Roche Registration GmBH, 2022). This medication is reimbursed by the HSE under the High-Tech Drug Scheme and must be stored in a refrigerator at 2 to 8 °C. The dosing assumptions and calculations used for this comparator are detailed further in section 4.

3. Clinical evidence

3.1. Study Design

The global Phase 3 clinical trial program for Vadadustat in the treatment of anaemia in dialysis-dependent chronic kidney disease (DD-CKD) was conducted under the INNO2VATE program. (K.-U. Eckardt *et al.*, 2021). This comprised of two pivotal Phase 3 trials designed to evaluate the safety and efficacy of Vadadustat compared to Darbepoetin alfa, a standard erythropoiesis-stimulating agent (ESA). Importantly, this trial was divided into two studies given the difference in patient characteristics among a population of dialysis patients: those who had recently started dialysis versus those who had been on dialysis long term.

The first study, the Incident DD-CKD trial (NCT02865850), enrolled 369 patients who had recently initiated dialysis within the previous 16 weeks at the point of screening and had limited prior exposure to ESA therapy or no more than two doses of a short-acting ESA (e.g., epoetin alfa) in the 8 weeks prior to screening. The second study, the Prevalent DD-CKD trial (NCT02892149), enrolled 3,554 patients who were already receiving maintenance dialysis with active ESA treatment. Both trials were Phase 3, randomised, open-label, sponsor-blind, active-controlled studies designed to assess the cardiovascular safety and haematological efficacy of Vadadustat for the long-term treatment of anaemia in DD-CKD patients.

The primary safety endpoint of both trials was time to first major adverse cardiovascular event (MACE), which included a composite measure of all-cause mortality, non-fatal myocardial infarction (MI) and non-fatal stroke. The primary efficacy endpoint was the change in haemoglobin (Hb) levels from baseline (mean pre-treatment Hb concentration) to the primary evaluation period (mean Hb concentration Weeks 24–36). The trials followed a structured evaluation process, beginning with a correction/conversion period (Weeks 0–23), followed by a maintenance period (Weeks 24–52) and concluding with a long-term treatment and safety follow-up period. Both INNO2VATE trials have been completed. The program was event driven and study completion was determined upon when 631 MACE events have accrued over the two DD-CKD studies.

Table 1.1: INNO2VATE-Correction DD-CKD-trial

clinicaltrials.gov identifier: NCT02865850	**Trial design:** A multicentre, randomised, open-label, sponsor blind, active-controlled Phase 3 study	
Treatments	**Key inclusion and exclusion criteria**	**Endpoints**
Intervention: [Vadadustat (AKB-6548), The starting dose for Vadadustat was 300 mg once daily (two 150 mg tablets), with doses of 150, 300, 450 and 600 mg daily available for dose titration during the study after an initial 4-week fixed-dose period, adjusted to maintain Hb levels in the target range: US: 10–11 g/dL Non-US: 10–12 g/dL, administered once daily, ≥36 weeks, **Stopping Rules:** Study medication must be permanently discontinued if a patient meets one of the criteria: • ALT or AST >3x Upper Limit of Normal (ULN) and total bilirubin >2x ULN • ALT or AST >3x ULN and INR >1.5 • ALT or AST >8x ULN • ALT or AST remains >5x ULN over 2 weeks* • ALT or AST >3x ULN with symptoms (e.g., fatigue, nausea, vomiting, right upper quadrant pain, fever, rash) or eosinophilia * Re-challenge generally should be avoided with ALT or AST >5x ULN unless there are no other good therapeutic options [n=181] Control: [Darbepoetin alfa IV/SC (starting dose based on the approved local product label), Darbepoetin alfa dosing is independent of the visit schedule, and the dosing schedule may shift per local standard-of-care and country specific dosing guidelines duration, **Stopping rules:** Study medication must be permanently discontinued if a subject meets one of the criteria: • ALT or AST >3x Upper Limit of	**Inclusion Criteria:** Subjects must meet the following inclusion criteria: 1. ≥18 years of age. 2. Initiated chronic maintenance dialysis (either peritoneal or haemodialysis) for end-stage kidney disease within 16 weeks prior to Screening. 3. Mean screening HGB <10.0 g/dL as determined by the average of 2 HGB values measured by the central laboratory during Screening. 4. Serum ferritin ≥100 ng/mL and transferrin saturation (TSAT) ≥20% at Screening. 5. Folate and vitamin B12 measurements ≥ lower limit of normal at Screening. 6. Understands the procedures and requirements of the study and provides written informed consent and authorisation for protected health information disclosure. **Exclusion Criteria:** Subjects must meet none of the following exclusion criteria: 1. Anaemia due to a cause other than CKD or subjects with active bleeding or recent blood loss. 2. Subjects with sickle cell disease, myelodysplastic syndromes, bone marrow fibrosis, hematologic malignancy, myeloma, haemolytic anaemia, thalassemia, or pure red cell aplasia. 3. Received more than 1 dose of long-acting (e.g., darbepoetin alfa, methoxy polyethylene glycol-epoetin beta [Mircera, C.E.R.A.]) or 2 doses of short-acting erythropoiesis-stimulating agent (ESA) (e.g., recombinant human erythropoietin [rHuEPO]) within 8 weeks prior to Screening. Patients may not receive any ESA during the screening period. 4. Patients may not receive any red blood cell transfusions during the Screening period. 5. Anticipated to recover adequate kidney function to no longer require dialysis. 6. Aspartate aminotransferase (AST)/serum	**Primary endpoint**: The primary endpoint used to assess the efficacy objective will be the mean change in Hb between Baseline (mean pretreatment Hb) and the primary evaluation period (mean Hb from Weeks 24 to 36). **Key secondary efficacy endpoint(s):** Average change in Hb value between Baseline (mean pretreatment Hb) and the secondary evaluation period (Weeks 40 to 52). Proportion of patients with Hb values within the target range during the primary evaluation period (Weeks 24 to 36). Proportion of patients with Hb values within the target range during the secondary evaluation period (Weeks 40-52). **Primary Safety Endpoint:** Time from the first dose date to the first MACE (all-cause mortality, nonfatal myocardial infarction, or nonfatal stroke) **Key Secondary Safety Endpoint:** Expanded MACE (MACE plus hospitalisation for either heart failure or a thromboembolic event, excluding vascular access failure). Time to first occurrence of a composite of death from cardiovascular causes, nonfatal myocardial infarction or nonfatal stroke. Time to death from cardiovascular causes. Time to death from any cause Health-related Quality of Life endpoint(s): Not explicitly recorded

Normal (ULN) and total bilirubin >2x ULN • ALT or AST >3x ULN and INR >1.5 • ALT or AST >8x ULN • ALT or AST remains >5x ULN over 2 weeks* • ALT or AST >3x ULN with symptoms (e.g., fatigue, nausea, vomiting, right upper quadrant pain, fever, rash) or eosinophilia * Re-challenge generally should be avoided with ALT or AST >5x ULN unless there are no other good therapeutic options etc] [n=188]	glutamic oxaloacetic transaminase (SGOT), alanine aminotransferase (ALT)/serum glutamic pyruvic transaminase (SGPT), or total bilirubin >2.0 x upper limit of normal (ULN) at or during Screening. Subjects with a history of Gilbert's syndrome are not excluded. 7. Uncontrolled hypertension (defined as confirmed pre-dialysis systolic blood pressure [BP] >190 mmHg or diastolic BP >110 mmHg at rest) at or during Screening. 8. Severe heart failure at or during Screening (New York Heart Association Class IV). 9. Acute coronary syndrome (hospitalisation for unstable angina, myocardial infarction); surgical or percutaneous intervention for coronary, cerebrovascular, or peripheral artery disease (aortic or lower extremity); surgical or percutaneous valvular replacement or repair; sustained ventricular tachycardia; hospitalisation for CHF; or stroke within 12 weeks prior to or during Screening. 10. History of active malignancy within 2 years prior to or during Screening, Protocol No. AKB-6548-CI-0016 Akebia — Company Confidential Page 11 of 78 except for treated basal cell carcinoma of skin, curatively resected squamous cell carcinoma of skin, cervical carcinoma in situ, or resected benign colonic polyps. 11. History of deep vein thrombosis (DVT) or pulmonary embolism (PE) requiring active treatment within 8 weeks prior to or during Screening. 12. History of hemosiderosis or hemochromatosis. 13. History of prior organ transplantation or scheduled organ transplant (patients on the kidney transplant wait-list or with a history of failed kidney transplant are not excluded), or prior hematopoietic stem cell or bone marrow transplant (corneal transplants and stem cell therapy for knee arthritis are not excluded). 14. Hypersensitivity to Vadadustat, darbepoetin alfa or any of their excipients. 15. Use of an investigational medication or participation in an investigational study within 30 days or 5 half-lives of the investigational medication (whichever is longer), prior to or during Screening. 16. Previous participation in this study or previous participation in a study with another hypoxia-inducible factor prolyl-hydroxylase	

	inhibitor (HIF-PHI) other than Vadadustat. 17. Females who are pregnant or breastfeeding. Women of childbearing potential who are unable or unwilling to use an acceptable method of contraception. 18. Non-vasectomised male subjects who are unable or unwilling to use an acceptable method of contraception. 19. Any other reason, which in the opinion of the Investigator, would make the patient not suitable for participation in the study.	

Akebia Therapeutics. Phase 3, Randomized, Open-Label, Active-Controlled Study Evaluating the Efficacy and Safety of Oral Vadadustat for the Correction or Maintenance Treatment of Anaemia in Subjects With Incident Dialysis-Dependent Chronic Kidney Disease (DD-CKD) (INNO2VATE - CORRECTION/CONVERSION) [Internet]. clinicaltrials.gov. 2022. Available from: https://clinicaltrials.gov/study/NCT02865850

Table 2.2 INNO2VATE – Prevalent-DD-CKD

Clinical Trials Identifier: NCT02892149	***Trial Design:*** *Phase 3, multicentre, open-label, sponsor blind, randomised controlled trial*	
Treatments	**Inclusion & Exclusion Criteria**	**Endpoints**
Intervention: Vadadustat (oral tablets) dose starting at 300mg OD with flexible titration according to Hb. Titration: additional 150mg to 300mg per day up to a maximum dose of 600mg per day Duration: >/= 36 weeks **Control:** Darbepoetin alfa (S/C or IV) dosed according to prior dosing and local label/guidelines with schedule according to local guidelines and response. **Monitoring:** Hb measured via HaemoCue point-of-care delivery systems. Hb measured every 2 weeks from week 0-12; every 4 weeks from weeks 12-52 and monthly from weeks 53 until study end **Iron Supplementation:** IV, oral or intradialytic administration was	**Inclusion Criteria:** 1. >18 years of age 2. Diagnosis of anaemia of DD-CKD 3. Understands procedures & requirements of study including completion of written consent 4. Serum ferritin >100ng/ml and transferring saturation >20% 5. B12 & Folate levels within normal range 6. Has not received RBC transfusion within 8 weeks prior to randomisation 7. Received maintenance dialysis (peritoneal or haemodialysis) for a minimum of 12 weeks prior to screening 8. Currently receiving any form of ESA therapy with dose within 6 weeks prior to screening 9. Hb 8-11 g/dL (USA) or 9-12 g/dL (non-USA) **Exclusion Criteria** 1. Anaemia secondary to any other cause including any current active bleeding or recent blood loss 2. Uncontrolled Hypertension 3. Recent cardiovascular event e.g. MI; surgical or percutaneous intervention for coronary, cerebrovascular or peripheral artery disease; any valvular replacement or repair; sustained VT or stroke within 12 weeks prior to or during screening 4. AST, ALT or Bilirubin >2.0 times upper limit of normal; Patients with Gilbert's syndrome not excluded	**Primary Efficacy Endpoint:** Change in average Hb between baseline and primary evaluation period (mean Hb concentration in weeks 24-36) **Key Secondary Efficacy Endpoint:** Change in average Hb between baseline and secondary evaluation period (mean Hb concentration in weeks 40-52) **Primary Safety Endpoint:** Time to first MACE (all-cause mortality, non-fatal MI, non-fatal stroke) **Key Secondary Safety Endpoints:** Time to first expanded MACE: MACE plus hospitalisation for heart failure or thromboembolic event, excluding vascular access failure. Time to cardiovascular mortality, non-fatal MI and non-fatal stroke Time to cardiovascular mortality Time to all-cause mortality

recommended to maintain serum ferritin concentration of at least 100ng/ml or transferrin saturation of at least 20%.	5. History of active malignancy within 2 years excepts BCC or resected SCC 6. History of PE or DVT within 12 weeks prior to randomisation 7. History of hemosiderosis or haemochromatosis 8. History of prior solid organ, haematopoietic stem cell or bone marrow transplant. Patients with prior failed kidney transplant not excluded. 9. Hypersensitivity to Vadadustat, darbepoetin alfa or any derivatives 10. Use of investigational or participation of investigational study within 30 days or 5 half-lives of the investigational drug treatment 11. Females who are pregnant, breastfeeding or childbearing potential and unable/unwilling to take contraception 12. Non-vasectomised males who are unable/unwilling to use an acceptable form of contraception 13. Any other reason, in opinion of investigator, would make patient not suitable for participation in study	

3.2. Clinical efficacy

3.2.1. INNO2VATE – Prevalent-DD-CKD trial

A total of 3554 patients were randomised (n=1777 Vadadustat vs. n=1777 Darbepoetin alfa) with 61.2% of patients residing in USA and remainder within Europe, South America and Asia. Baseline characteristics were well-balanced between the intervention and control groups. The mean age of the Vadadustat group was 57.9 years (vs 58.4 years for Darbepoetin alfa) and 44.3% were female (vs 43.5% for Darbepoetin alfa). Majority of individuals within the trial were ethnically white, 63.9% Vadadustat group vs 61.7% for Darbepoetin group. Mean haemoglobin was 10.25 for Vadadustat compared to 10.23 for Darbepoetin group, with majority of patients on haemodialysis (1652 vs 1633). The mean number of years on dialysis was 4.00 for Vadadustat group (vs 3.94 for Darbepoetin alfa). The prevalence of cardiovascular disease (CVD) at baseline was comparable between the two treatment groups, with 48.8% of patients in the Vadadustat group and 52.4% in the Darbepoetin Alfa group having a history of CVD. The mean baseline haemoglobin concentration was slightly higher in the Vadadustat group at 10.6 g/dL compared to 10.2 g/dL in the Darbepoetin Alfa group. A slightly higher proportion of patients in the Vadadustat group had diabetes mellitus compared to the Darbepoetin Alfa group (54.6% vs. 56.2%).

3.2.2. INNO2VATE – Incident-DD-CKD trial

A total of 369 patients were randomised with 52.8% of patients residing in the USA and the remained in Europe. (n=188 Vadadustat vs. n=188 Darbepoetin Alfa). The mean age for the Vadadustat group was 56.5 years(vs 55.6 years for Darbepoetin) and 40.9% were female(vs 39.1% for Darbepoetin Alfa).The majority of patients were ethnically white(71.3% Vadadustat group vs. 76.1% for Darbepoetin Alfa group), followed by black patients(21.0% Vadadustat group vs 18.6% Darbepoetin Alfa group)and Asian patients(6.6% Vadadustat group vs 4.3% Darbepoetin Alfa group) (K. U. Eckardt *et al.*, 2021). The prevalence of cardiovascular disease at baseline was similar between the two treatment groups, with 38.1% of patients in the Vadadustat group and 38.8% in the Darbepoetin alfa group having a history of cardiovascular disease. The mean baseline haemoglobin concentration was slightly higher in the Vadadustat group at 9.4 g/dL compared to 9.2 g/dL in the Darbepoetin alfa group. In terms of dialysis modality, most patients in both groups were receiving in-centre haemodialysis, accounting for 87.3% of Vadadustat-treated patients and 90.9% of those receiving Darbepoetin alfa (K. U. Eckardt *et al.*, 2021). A smaller proportion of patients were undergoing peritoneal dialysis, with 12.2% in the Vadadustat group and 8.6% in the Darbepoetin alfa group. However, a higher percentage of patients in the Vadadustat group was observed to diabetes mellitus compared to the Darbepoetin Alfa group (58.0% vs 51.1%) (K. U. Eckardt *et al.*, 2021) which may be a clinically relevant imbalance considering the increased cardiovascular risk associated with diabetes mellitus (National Institute of Diabetes and Digestive and Kidney Diseases (NIDDKD), 2021). This imbalance may have influenced the incidence of MACE events in the Vadadustat group.

Table 2.1: INNO2VATE Correction clinical outcomes

Date of *final* analysis: 2020-03-30		
Outcome	**Vadadustat**	**Darbepoetin alfa**
Change from Baseline Hb to Average over Primary Efficacy Period (week 24 to 36) Least Squares Mean(g/dl)	**1.26** ***[±0.11]***	**1.58** ***[±0.11]***
Median Time to First MACE (weeks)	26.21	46.64
Pooled Analysis: AKB-6548-CI-0016 (NCT02865850) and AKB-6548-CI-0017 (NCT02892149)	46.14	47.00
Change from Baseline Hb o Average over Secondary Efficacy Period (weeks 40 to 52) Least Squares Mean (g/dL) [±Standard Error]	**1.42** ***[±0.13]***	**1.50** ***[±0.14]***
Median Time to first expanded MACE (weeks)	**27.14**	**47.00**
Pooled Analysis: AKB-6548-CI-0016 (NCT02865850) and AKB-6548-CI-0017 (NCT02892149)	42.07	45.29
Median Time to First Cardiovascular MACE	**18.50**	**54.07**
Pooled Analysis: AKB-6548-CI-0016 (NCT02865850) and AKB-6548-CI-0017 (NCT02892149)	43.29	45.79
Median Time to cardiovascular mortality (weeks)	**22.00**	**59.14**
Pooled Analysis: AKB-6548-CI-0016 (NCT02865850) and AKB-6548-CI-0017 (NCT02892149)	43.71	49.29
Median Time to all-cause mortality (weeks)	**31.71**	**45.36**
Pooled Analysis: AKB-6548-CI-0016 (NCT02865850) and AKB-6548-CI-0017 (NCT02892149)	50.00	49.57
No health-related quality of life endpoints was determined in this trial		

g/dL = grams per decilitre; MACE = Major Adverse Cardiovascular Event

Footnote: *Data for all outcomes in this table are from the NIH. Efficacy and Safety Study to Evaluate Vadadustat for the Correction or Maintenance Treatment of Anaemia in Participants With Incident Dialysis-dependent Chronic Kidney Disease (DD-CKD) [Internet]. Clinicaltrials.gov. 2025 [cited 2025 Mar 10]. Available from: https://clinicaltrials.gov/study/NCT02865850?*

Table 3.2: INNO2VATE Prevalent clinical outcomes

Outcome	*Vadadustat*	*Darbepoetin Alfa*
Change from Baseline Hb to Average over Primary Efficacy Period (weeks 24 to 36) Least Squares Mean (g/dL) [±Standard Error]	***0.19 [±0.03]***	***0.36 [±0.03]***
Median Time to First MACE (weeks)	***48.86***	***48.00***
Change from Baseline Hb o Average over Secondary Efficacy Period (weeks 40 to 52) Least Squares Mean (g/dL) [±Standard Error]	*0.23 [±0.04]*	*0.41 [±0.03]*
Median Time to first expanded MACE (weeks)	*44.57*	*43.57*
Median Time to cardiovascular mortality (weeks)	*46.14*	*47.64*
Median Time to all-cause mortality (weeks)	*50.79*	*50.43*
No health-related quality of life endpoints were determined in this trial		

g/dL = grams per decilitre; MACE = Major Adverse Cardiovascular Event

Footnote: *Data for all outcomes in this table are from the NIH. Efficacy and Safety Study to Evaluate Vadadustat for the Maintenance Treatment of Anaemia in Participants With Dialysis-dependent Chronic Kidney Disease (DD-CKD) [Internet]. Clinicaltrials.gov. 2025 [cited 2025 Mar 10]. Available from: https://clinicaltrials.gov/study/NCT02892149*

3.2.3. Summary of Other Clinical Evidence of Relevance

A Phase 3, randomised, open-label, active-controlled study conducted in Japan assessed the efficacy and safety of Vadadustat compared to darbepoetin alfa in haemodialysis patients with anaemia due to chronic kidney disease (CKD) (Nangaku *et al.*, 2021). Over a 52-week treatment period, Vadadustat demonstrated noninferiority to darbepoetin alfa in maintaining haemoglobin (Hb) levels within the target range of 10.0–12.0 g/dL, with mean Hb values remaining comparable between treatment groups. Subgroup analyses further confirmed consistent efficacy across patients with varying dialysis durations and anaemia histories. The overall safety profile of Vadadustat was similar to that of darbepoetin alfa with commonly reported adverse events, including nasopharyngitis, diarrhoea, and shunt stenosis, occurring at comparable rates. Notably, a slightly higher incidence of retinal haemorrhage was observed in the Vadadustat group (9.9% vs. 6.2%) though these events were predominantly mild and not directly attributable to the drug (Nangaku *et al.*, 2021). Cardiovascular adverse events and thromboembolic complications were reported at similar frequencies in both treatment arms, reinforcing the cardiovascular safety of Vadadustat in this population.

3.3. Clinical safety

The most reported adverse events across both INNO2VATE trials were hypertension, diarrhoea, pneumonia, hyperkalaemia, fluid overload, headache, hypotension, nausea, vomiting, urinary tract infections, and dialysis-related complications. Variations were observed between the two studies. In the Incident DD-CKD Trial, the most frequent adverse events were hypertension (16.2% in the Vadadustat group vs. 12.9% in the Darbepoetin alfa group), diarrhoea (10.1% vs. 9.7%), and pneumonia (7.3% vs. 8.1%). In the Prevalent DD-CKD Trial, diarrhoea (13.0% vs. 10.1%) and pneumonia (11.0% vs. 9.7%) were more common in Vadadustat users, whereas hypertension was more frequent in Darbepoetin alfa users (10.6% vs. 13.8%) (K.-U. Eckardt *et al.*, 2021).

The incidence of Grade 3-5 serious adverse events (SAEs) was high in both studies. In the Incident DD-CKD Trial, SAEs occurred in 49.7% of Vadadustat patients vs. 56.5% of Darbepoetin alfa patients, with adverse events leading to death in 8.4% vs. 9.7%, respectively. The Prevalent DD-CKD Trial showed a higher overall SAE incidence, affecting 55.0% of Vadadustat users vs. 58.3% of Darbepoetin alfa users, with death rates of 15.0% vs. 15.6%, respectively (K.-U. Eckardt *et al.*, 2021). These findings suggest that Vadadustat has a comparable or slightly improved safety profile relative to Darbepoetin alfa in dialysis-dependent CKD patients.

The primary safety endpoint with Vadadustat was the occurrence of a major adverse cardiovascular events (MACE), which include death from any cause, nonfatal myocardial infarction, or nonfatal stroke. Across both trials, the first occurrence of a MACE was reported in 18.2% of Vadadustat users and 19.3% of Darbepoetin alfa users. The hazard ratio (HR) for MACE was 0.96 (95% confidence interval [CI]: 0.83–1.11), indicating no statistically significant difference in cardiovascular risk between treatment groups. Vadadustat was determined to be noninferior to Darbepoetin Alfa, based on the prespecified noninferiority margin of 1.25 (per FDA) and 1.30 (per EMA) (Akebia Therapeutics, 2022) . A key finding from the incident dialysis population was that patients receiving Vadadustat experienced their first MACE event sooner than those treated with Darbepoetin Alfa. This observation suggests a potentially higher early cardiovascular risk in incident dialysis patients treated with Vadadustat, despite the overall MACE rate being comparable between the two treatment groups. This finding suggests the need for closer cardiovascular monitoring in patients newly initiated on dialysis when prescribing Vadadustat, although the overall cardiovascular safety profile remains consistent with that of Darbepoetin Alfa in the broader dialysis-dependent CKD population. Patients with signs and symptoms of serious adverse cardiovascular reactions or stroke should be undergo prompt evaluation and be treated according to standard of care. The decision to interrupt or discontinue treatment should be guided on a benefit-

risk consideration for the individual patient (Medice Arzneimittel Pütter GmbH & Co. KG, 2024).

Additionally, the expanded MACE, which includes hospitalisation for heart failure or thromboembolic events, occurred in 21.6% of Vadadustat users vs. 23.0% of Darbepoetin alfa users. These findings indicate that the overall incidence of expanded MACE events was comparable between the two treatment groups, reinforcing the noninferiority of Vadadustat in terms of cardiovascular safety. However, significant differences in hypertension incidence were noted between the two trials, suggesting potential variations in the hemodynamic effects of Vadadustat depending on dialysis duration and prior erythropoiesis-stimulating agent (ESA) exposure. In the Incident Dialysis-Dependent Chronic Kidney Disease (DD-CKD) trial, Vadadustat was associated with a higher incidence of hypertension (16.2%) compared to Darbepoetin Alfa (12.9%), whereas in the Prevalent DD-CKD trial, the incidence was lower among Vadadustat-treated patients (10.6%) relative to those receiving Darbepoetin Alfa (13.8%) (K.-U. Eckardt *et al.*, 2021). These findings highlight the need for careful patient stratification when initiating Vadadustat particularly in patients newly undergoing dialysis, where blood pressure monitoring may be warranted to mitigate potential hypertensive effects. Patients presenting with signs or symptoms of thromboembolic events should be promptly assessed and managed according to standard medical guidelines. The decision to pause or discontinue treatment should be based on an individualised evaluation of the benefits and risks for each patient (Medice Arzneimittel Pütter GmbH & Co. KG, 2024).

Furthermore, across both the incident DD-CKD and prevalent DD-CKD trial, no significant differences were observed between Vadadustat and Darbepoetin alfa in terms of serum ferritin, hepcidin, or transferrin saturation levels indicating a comparable impact on iron metabolism. However, patients receiving Vadadustat required more frequent rescue therapy with erythropoiesis-stimulating agents (ESAs) compared to those on Darbepoetin alfa, particularly in the Incident DD-CKD Trial, where 20.4% of Vadadustat-treated patients required ESA rescue therapy versus 16.0% in the Darbepoetin alfa group. In the Prevalent DD-CKD Trial, the trend persisted with 27.6% of Vadadustat users requiring ESA rescue therapy compared to 30.2% of Darbepoetin alfa user (K.-U. Eckardt *et al.*, 2021). The increased need for rescue therapy in Vadadustat-treated patients, especially in newly dialysis-dependent patients indicates that its haemoglobin response may be slower compared to Darbepoetin alfa. Clinicians may need to monitor patients closely and be prepared to use ESA therapy to maintain stable haemoglobin levels particularly in the early stages of treatment.

4. Comparative costs

The following table details the assumptions made in reference to the drug cost and dosing regimen choice. The rationale column describes the back of the envelope calculations using a base case weight and references from where cost was determined. These assumptions support the final calculations for comparative costs detailed in Table 5.

Table 3: Assumptions and rationale for comparative costs and dosing for Vadadustat & Comparators

Assumptions	*Rationale*
Oral Dose of Vadadustat 300 milligrams once daily	Recommended starting dose for drug as per SmPC and corresponds with average dose throughout INNO2VATE trial.
Subcutaneous Dose of Darbepoetin Alfa 35 micrograms per week	Based on dosing schedule for DD-CKD patients at 0.45mcg/kg/week as per product SmPC (Amgen Europe, 2023). Average weight for western European adult determined as 76.5kg (Worlddata.org, 2022). This leads to weekly dose of 35mcg. This corresponds with average dose administered throughout INNO2VATE trial.
Subcutaneous Dose of Epoetin Beta 8262 IU per week	Based on mean Epoetin Beta dosing of 108IU/kg/week determined from GAIN study (Rath *et al.*, 2009) and applied to average weight for western European adult determined as 76.5kg (Worlddata.org, 2022). This results in a weekly dose of 8262 IU.
Patients on Darbepoetin Alfa receive subcutaneous preparation	Dosing calculations based on community pharmacy dispensed medication rather than hospital medication which is more likely to be IV preparation.
Vadadustat Cost	Cost determined from NICE TA1035 (National Institute for Health and Care Excellence (NICE), 2025a)documented price to wholesaler (£148.59) per 28-tablet 150mg pack. This was converted into Irish Euros using AIB currency converter (€179.97). Two packs are required per 28-day treatment cycle.
Darbepoetin Alfa Cost	Cost determined from HSE price data (Health Service Executive (HSE), 2022)for a single 20mcg pre-filled pen (€36.63). Two pre-filled pens are required for dose of 35mcg with 5mcg wastage. This is the minimum cost to achieve the required dose. A pack of four pre-filled pens at a cost of €135.68 per 14-day treatment cycle was used for our analysis.

Epoetin Beta Cost	To achieve a dose of 8262 IU per week, dose must be administered via three different sized injections: 5000 IU (€34.19), 3000 IU (€20.35) and 500 IU (€3.295) with wastage. Costs were determined from HSE Price Realignment data (Health Service Executive (HSE), 2024) where price for single pre-filled syringe was determined from 6-pack prices. The total result is a cost of €57.80 per week or €115.60 per 14-day treatment cycle requiring two packs. We have assumed a hypothetical pack containing one each of the required syringes.
Additional Costs	Additional costs for drug monitoring, blood tests for Hb levels, are not included as these would be considered a patient cost.
Mcg= Micrograms; kg=kilograms; IU= International Units; HSE= Health Service Executive; SmPC= Summary of Product Characteristics	

Table 4: Total comparative drug cost to the HSE for the intervention and comparator(s)

	Total cost per patient per year*		
	Including VAT	Excluding VAT	Source¢
Vafseo (Vadadustat)§	€4,711.52‡	€4,711.52	NICE TA1035; converted to Irish Euros (2025) using AIB currency converter
Aranesp SureClick Pen (Darbepoetin Alfa) †	€5,128.04¶	€4,248.76	HSE Price Realignment (2022)
Neo Recormon Pre-filled Syringe (Epoetin Beta)|	€4,479.28¶	€3,730.12	HSE Price Realignment (2024)

*Including all relevant fees and rebates (rate of Framework Agreement rebate –9%)
‡ 0% rate of VAT on oral medicines
§ Based on starting dose of 300mg per day; based on 28-tablet 150mg pack size; price per pack = £148.59
† Based on 0.45mcg/kg dose for 76.5kg adult per week (worlddata.org average weight for western Europe); Rounded to 35mcg dose per week (2x20mcg Pre-filled pen with 5mcg drug wastage as 15mcg pre-filled pen not available; price per pre-filled pen = €36.63); this corresponds to dosing in Eckardt et al 2021.
¶ 23% rate of VAT on non-oral medicines
|Based on 108IU/kg dose for 76.5kg adult per week; 8262 IU/week (1x5000IU pre-filled syringe, 1x3000IU pre-filled syringe & 1x500IU pre-filled syringe with drug wastage); dosing determined from Rath et al 2009 and fits with regime in SmPC
¢Full source reference and access link available in reference section

5. Budget Impact Analysis

The budget impact model for Vadadustat in the Irish healthcare system estimates a total drug cost of €12.65 million over five years. However, its introduction is projected to generate net savings of €1.12 million, as it reduces reliance on existing treatments like Darbepoetin Alfa and Epoetin Beta, whose combined costs without Vadadustat would reach €36.15 million (including VAT). With Vadadustat available, the comparator costs drop to €22.38 million, leading to substantial cost reductions. The net drug budget impact (including VAT) is -€1,121,794, while the net drug budget impact (excluding VAT) is +€1,233,700, reflecting potential cost shifts. The full yearly budget impact analysis for Vadadustat within the Irish Healthcare systems is summarised in Table 6. The following paragraphs outline how the eligible population was determined including mortality rates and discontinuation rates; market share assessment and postulations about wider health service impacts.

5.1. Eligible Population

The prevalence and incidence of ESKD in Ireland were derived from the National Renal Office (NRO) Renal Services Report 2024. Based on population estimates the total adult (18+) population in Ireland is projected to be 4,236,154 in 2025. The total prevalence of treated ESKD patients in 2023 was 5,190, which aligns with an estimated 0.10% prevalence of ESKD among adults (Mellotte, 2024). Among patients with ESKD, 47.9% require dialysis, comprising 42.2% receiving in-centre haemodialysis and 5.7% receiving peritoneal/home dialysis (9). For consistency, this calculation was rounded to 48% in the budget impact analysis. The annual incidence rate of ESKD was estimated at 2.2% in 2023 (Mellotte, 2024). This incidence rate was applied to estimate the expected yearly increase in patients requiring dialysis and subsequent anaemia treatment. The incident patient population reflects the projected annual growth in dialysis-dependent CKD patients who develop anaemia and require ESA therapy. Moreover, according to a 2024 study by Hashmi et al, at least 90% of dialysis-dependent patients develop anaemia (Hashmi, Aeddula and Shaikh, 2024). Accordingly, the model assumes that 90% of dialysis-dependent patients will receive ESA therapy.

A 16.5% annual mortality rate was applied to the dialysis-dependent population each year with calculations based on multiplication of the mortality rate by the total dialysis-dependent patient population. This method estimates the annual reduction in the number of patients eligible for the intervention over five years. Moreover, the net number of patients was calculated by subtracting the annual mortality rate from the total number of patients with the condition. As 100% of the net population remained eligible for the new intervention under

license, the net number of patients was carried over annually and continued to represent the eligible patient population over the five-year period.

5.2. Market share without Intervention

In the absence of Vadadustat, the market share is dominated by existing treatments, primarily Darbepoetin Alfa and Epoetin Beta. Darbepoetin Alfa remains the preferred choice, capturing 80% of the market share as determined from a NICE budget impact model for a similar HIF-PIF (National Institute for Health and Care Excellence, 2022). In comparison to this NICE budget impact model, in Ireland there is no availability of other Epoetin derivatives covered by this market authorisation indication, so an assumption that Epoetin Beta makes up the remainder of the market share of 20% is made.

The expected annual growth rate for Darbepoetin Alfa is projected at 5.2% as per the expected earnings report (Spherix Global Insights, 2023a). However, taking a more conservative approach, a growth rate of 4% is assumed, declining and stabilising at 3% from Year 3 onwards due to the potential introduction of novel drug treatments.

5.3. Market share with Intervention:

According to market reports, the introduction of Vadadustat for the treatment of anaemia is expected to lead to its adoption by approximately 25% of patients in the first year (Spherix Global Insights, 2023b). The anticipated year-on-year growth is projected at 15% in Year 2, followed by 23% in Year 3 and 25% in subsequent years, resulting in a market share of 55.25% by Year 5. This projection aligns with the estimated market penetration outlined in the resource utilisation approval by NICE for Vadadustat (National Institute for Health and Care Excellence (NICE), 2025b). The year-on-year growth of Darbepoetin Alfa and Epoetin Beta is assumed to follow the same trajectory as in the scenario without intervention.

5.4. Discontinuation rates:

The discontinuation rates for Vadadustat and Darbepoetin Alfa remain constant over the five-year period at 5.10% and 1.10%, respectively, using data from the INNO2VATE trial (K.-U. Eckardt *et al.*, 2021). These rates are derived from clinical data on the safety and efficacy of Vadadustat for anaemia in patients undergoing dialysis. Additionally, a discontinuation rate of 9.50% is considered for Epoetin Beta (Locatelli *et al.*, 2002).

5.5. Potential Wider Healthcare & Societal Impact

The benefits of Vadadustat extend beyond its clinical effects. While challenging to quantify within the constraints of this rapid review, its lack of refrigeration requirements and reduced waste compared to single-use injection pens offer notable sustainability advantages. There is also potential cost savings from the proportion of patients are unable to self-inject and who require clinical staff to attend to administer the comparator ESAs. Additionally, ease of administration may support compliance with treatment for anaemia in DD-CKD.

Table 5: Drug-budget impact

Population	Year 1	Year 2	Year 3	Year 4	Year 5	5-year cumulative
Eligible population*	1528	1591	1620	1648	1676	8063
Proportion treated (%)	25.00%	28.75%	35.36%	44.20%	55.25%	
Treated population	382	457	573	729	926	3067
Gross drug-budget impact						
Including VAT	€1,576,150†	€1,887,409†	€2,362,95†	€3,005,877†	€3,821,260†	€12,653,654†
Excluding VAT	€1,576,150	€1,887,409	€2,362,958	€3,005,877	€3,821,260	€12,653,654
Net drug-budget impact						
Including VAT	-€119,747	-€154,655	-€204,617	-€274,698	-€368,078	-€1,121,794
Excluding VAT	€169,810	€194,251	€234,314	€286,431	€348,894	€1,233,700

* Based on a total population prevalence of 0.1% for ESKD, where almost half are deemed DD-CKD and majority of those have anaemia and are eligible for Vadadustat therapy
† Footnote indicating that there is a 0% rate of VAT on oral medicines

6. International HTA

Table 6: International HTA

Organisation	Recommendation/Status (date)
NICE, UK	Recommended as treatment option for symptomatic anaemia in DD-CKD (23/01/2025)
SMC, Scotland	Not currently under consideration
Japan	Recommended as treatment option for symptomatic anaemia in DD-CKD & non-dialysis dependent CKD (01/06/2020)
ZIN, Netherlands	Recommended as treatment option for symptomatic anaemia in DD-CKD (03/12/2024)
AIHTA, Austria	No information available
CADTH, Canada	
PBAC, Australia	Recommended as treatment option for symptomatic anaemia in DD-CKD (25/09/2023)
ICER, US	FDA approval as treatment option for symptomatic anaemia in DD-CKD, available on the market (01/01/2025)

7. References

Akebia Therapeutics (2022) *Efficacy and Safety Study to Evaluate Vadadustat for the Correction or Maintenance Treatment of Anaemia in Participants With Incident Dialysis-dependent Chronic Kidney Disease (DD-CKD)*, *Clinicaltrials.gov*. Available at: https://clinicaltrials.gov/study/NCT02865850?intr=inno2vate&cond=anaemia%20ckd&rank=2&tab=resultsN (Accessed: 10 March 2025).

Amgen Europe (2023) *Aranesp Darbepoetin Alfa | Summary of Product Characteristics*. Available at: https://www.medicines.ie/medicines/aranesp-sureclick-pen-31306/spc (Accessed: 12 March 2025).

Bello, A.K. *et al.* (2019) 'Status of care for end stage kidney disease in countries and regions worldwide: International cross sectional survey', *The BMJ*, 367. Available at: https://doi.org/10.1136/bmj.l5873.

Chung, E.Y. *et al.* (2023) 'Erythropoiesis-stimulating agents for anaemia in adults with chronic kidney disease: a network meta-analysis', *Cochrane Database of Systematic Reviews*, 2023(2). Available at: https://doi.org/10.1002/14651858.CD010590.pub3.

Eckardt, K.U. *et al.* (2021) 'Global Phase 3 programme of Vadadustat for treatment of anaemia of chronic kidney disease: Rationale, study design and baseline characteristics of dialysis-dependent patients in the INNO2VATE trials', *Nephrology Dialysis Transplantation*, 36(11). Available at: https://doi.org/10.1093/ndt/gfaa204.

Eckardt, K.-U. *et al.* (2021) 'Safety and Efficacy of Vadadustat for Anaemia in Patients Undergoing Dialysis', *New England Journal of Medicine*, 384(17). Available at: https://doi.org/10.1056/nejmoa2025956.

Egrie, J.C. and Browne, J.K. (2001) 'Development and characterization of novel erythropoiesis stimulating protein (NESP)', *British Journal of Cancer*, 84(SUPPL. 1). Available at: https://doi.org/10.1054/bjoc.2001.1746.

Ershler, W.B. *et al.* (2005) 'Economic burden of patients with anaemia in selected diseases', *Value in Health*, 8(6). Available at: https://doi.org/10.1111/j.1524-4733.2005.00058.x.

Gupta, N. and Wish, J.B. (2017) 'Hypoxia-Inducible Factor Prolyl Hydroxylase Inhibitors: A Potential New Treatment for Anaemia in Patients With CKD', *American Journal of Kidney Diseases*. Available at: https://doi.org/10.1053/j.ajkd.2016.12.011.

Hashmi, M., Aeddula, N. and Shaikh, H. (2024) *Anaemia of Chronic Kidney Disease* , *StatPearls [Internet]*. Available at: https://www.ncbi.nlm.nih.gov/books/NBK539871/ (Accessed: 12 March 2025).
Health Service Executive (HSE) (2022) *IPHA Price Reduction 2022: HSE Price Realignment File (Excel)*, *Corporate Pharmaceutical Unit Price Lists*. Available at: https://www.hse.ie/eng/about/who/cpu/ipha-price-reduction-2022/ (Accessed: 13 March 2025).

Health Service Executive (HSE) (2024) *IPHA Price Reduction 2024: HSE Price Realignment File (Excel)*, *Corporate Pharmaceutical Unit Price Lists*. Available at: https://www.hse.ie/eng/about/who/cpu/ipha-price-reduction-2024/ (Accessed: 13 March 2025).
Kaballo, M.A. *et al.* (2018) 'A comparative analysis of survival of patients on dialysis and after kidney transplantation', *Clinical Kidney Journal*, 11(3), pp. 389–393. Available at: https://doi.org/10.1093/ckj/sfx117.

Kalantar-Zadeh, K. *et al.* (2021) 'Chronic kidney disease', *The Lancet*. Available at: https://doi.org/10.1016/S0140-6736(21)00519-5.

Locatelli, F. *et al.* (2002) 'Once-weekly compared with three-times-weekly subcutaneous epoetin β: Results from a randomized, multicentre, therapeutic-equivalence study', *American Journal of Kidney Diseases*, 40(1), pp. 119–125. Available at: https://doi.org/10.1053/ajkd.2002.33920.

Locatelli, F. *et al.* (2008) 'Clinical practice guidelines for anaemia in chronic kidney disease: Problems and solutions. A position statement from Kidney Disease: Improving Global Outcomes (KDIGO)', in *Kidney International*, pp. 1237–1240. Available at: https://doi.org/10.1038/ki.2008.299.

McMurray, J.J.V. *et al.* (2012) 'Kidney disease: Improving global outcomes (KDIGO) anaemia work group. KDIGO clinical practice guideline for anaemia in chronic kidney disease', *Kidney International Supplements*. Available at: https://doi.org/10.1038/kisup.2012.37.

Medice Arzneimittel Pütter GmbH & Co. KG (2024) *Vafseo Vadadustat I SUMMARY OF PRODUCT CHARACTERISTICS*. Available at: https://ec.europa.eu/health/documents/community-register/2023/20231115161088/anx_161088_en.pdf (Accessed: 12 March 2025).

Mellotte, G. (2024) *National Renal Office: Renal Services Report 2024*. Available at: https://www.hse.ie/eng/about/who/cspd/ncps/renal/resources/nro-report-epidemiology-of-eskd-in-ireland-2024.pdf (Accessed: 12 March 2025).

Nangaku, M. *et al.* (2021) 'Efficacy and safety of Vadadustat compared with darbepoetin alfa in Japanese anaemic patients on haemodialysis: A Phase 3, multicentre, randomized, double-blind study', *Nephrology Dialysis Transplantation*, 36(9). Available at: https://doi.org/10.1093/ndt/gfab055.

National Institute for Health and Care Excellence (2022) *Roxadustat for treating symptomatic anaemia in chronic kidney disease*.

National Institute for Health and Care Excellence (NICE) (2021) *NICE guideline [NG203]: Chronic kidney disease: assessment and management*, *NICE Guidance*. Available at: https://www.nice.org.uk/guidance/ng203/chapter/Recommendations#managing-anaemia (Accessed: 12 March 2025).

National Institute for Health and Care Excellence (NICE) (2025a) *NICE Technology Appraisal [TA1035]: Vadadustat for treating symptomatic anaemia in adults having dialysis for chronic kidney disease*, *NICE Technology Appraisal* . Available at: https://www.nice.org.uk/guidance/ta1035/chapter/2-Information-about-vadadustat (Accessed: 13 March 2025).

National Institute for Health and Care Excellence (NICE) (2025b) *Vadadustat for treating symptomatic anaemia in adults having dialysis for chronic kidney disease*, *NICE Technology Appraisal* . Available at: https://www.nice.org.uk/guidance/ta1035/chapter/2-Information-about-vadadustat (Accessed: 13 March 2025).

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDKD) (2021) *Diabetes, Heart Disease, & Stroke*, *NIH NIDDKD Health Information* . Available at: https://www.niddk.nih.gov/health-information/diabetes/overview/preventing-problems/heart-disease-stroke (Accessed: 12 March 2025).

Ogieuhi, I.J. *et al.* (2024) 'Efficacy and safety of Vadadustat in the treatment of anaemia associated with chronic kidney disease: insights from clinical trials', *Discover Medicine*, 1(1), p. 147. Available at: https://doi.org/10.1007/s44337-024-00160-1.

Palmer, S.C. *et al.* (2014) 'Darbepoetin for the anaemia of chronic kidney disease', *Cochrane Database of Systematic Reviews*. Available at: https://doi.org/10.1002/14651858.CD009297.pub2.

Rath, T. *et al.* (2009) 'Effectiveness and safety of recombinant human erythropoietin beta in maintaining common haemoglobin targets in routine clinical practice in Europe: the GAIN study', *Current Medical Research and Opinion*, 25(4), pp. 961–970. Available at: https://doi.org/10.1185/03007990902784459.

Roche Registration GmBH (2022) *Epoetin Beta Neo Recormon | Summary of Product Characteristics*. Available at: https://www.medicines.ie/medicines/neorecormon-solution-for-injection-pre-filled-syringe-33005/spc (Accessed: 13 March 2025).

Spherix Global Insights (2023a) *FibroGen/Astellas' EVRENZO (roxadustat) Making In-Roads in European Market as In-Class Challenger, Akebia/Medice's VAFSEO (vadadustat), Prepares for Launch in Early 2024.*, *Nephrology Press Release* . Available at: https://www.spherixglobalinsights.com/fibrogen-astellas-evrenzo-roxadustat/ (Accessed: 12 March 2025).

Spherix Global Insights (2023b) *VAFSEO (vadadustat), Prepares for Launch in Early 2024*. Available at: https://www.spherixglobalinsights.com/fibrogen-astellas-evrenzo-roxadustat/ (Accessed: 12 March 2025).
Storring, P.L. *et al.* (1998) 'Epoetin alfa and beta differ in their erythropoietin isoform compositions and biological properties', *British Journal of Haematology*, 100(1). Available at: https://doi.org/10.1046/j.1365-2141.1998.00521.x.

Webster, A.C. *et al.* (2017) 'Chronic Kidney Disease', *The Lancet*. Available at: https://doi.org/10.1016/S0140-6736(16)32064-5.

Worlddata.org (2022) *Average height and weight by country*, *NCD Risk Factor Collaboration*. Available at: https://www.worlddata.info/average-bodyheight.php (Accessed: 3 March 2025).